

Release Outputs Required

Current required-output checklist for the July_04 engineering review flow.

Required Output	Owner	Release Criteria	Current State
Signed mechanical specification	Mechanical / system engineering	Envelope, datum, rails, keepouts, holes, connector exits, heatsink, and airflow confirmed.	Current Edge.Cuts approx. 305.0 mm x 260.5 mm; chassis constraints need confirmation.
Native schematic and schematic PDF/SVG	Electrical engineering	Dual NCE, TFLN PIC, RF/bias, HBM, PCIe/control/power blocks visible and aligned to netlist.	Readable overview exists; complete schematic/netlist source is still required.
Closed netlists	Electrical engineering	Generated from closed schematic and loaded into PCB with no unexplained differences.	214 schematic-parity warnings remain until authoritative source is imported.
PCB native project	PCB layout engineering	Updated KiCad project with 0 DRC errors and 0 unintended unconnected items.	Current DRC: 0 errors, 0 unconnected items, 82 library/footprint mismatch warnings.
Gerbers, drill, drill map, stackup, impedance notes	PCB layout / manufacturer CAM	Generated from reviewed PCB and approved by DFM.	Post-cleanup Gerber/drill review package generated; stackup and impedance require manufacturer confirmation.
Placement and assembly outputs	PCB layout / assembly engineering	XY placement, centroid, assembly drawings, polarity and DNP markings match signed BOM.	Placement/fiducials updated; BOM/DNP/populate owner signoff required.
Procurement BOM	Component / procurement engineering	MPN, value, package, quantity, source/AVL, DNP, alternates, revision/date signed.	Current July_04 BOM review list updated; AVL, quantity, substitutions, and parity require owner signoff.
Validation report	Electrical / PCB verification	DRC 0 errors, unconnected 0, schematic parity closed or signed-waived, ERC reviewed.	Current validation: 82 DRC warnings, 0 unconnected, 214 parity warnings.
DFM/DFA closure report	Manufacturer with engineering review	CAM, panelization, solder mask, silkscreen, drill, stackup, assembly and test risks closed.	DFM/DFA review should use post-cleanup Gerber/drill package.
Release manifest	Release engineering	All files listed with SHA256, revision, date, and owner approval.	Repository checksum manifest regenerated after this cleanup.